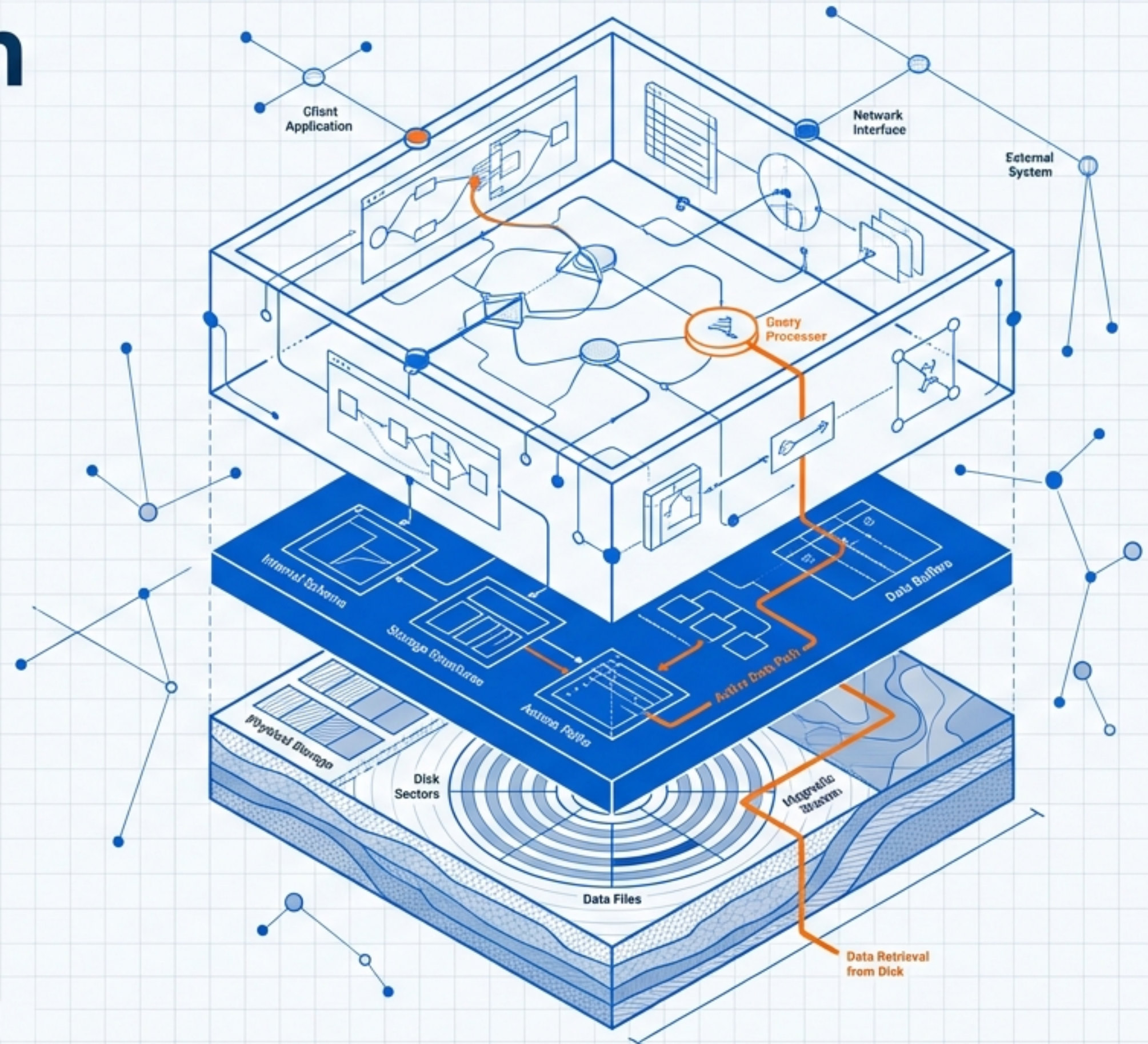


Database System Concepts and Architecture

A Structural Deconstruction of DBMS Fundamentals

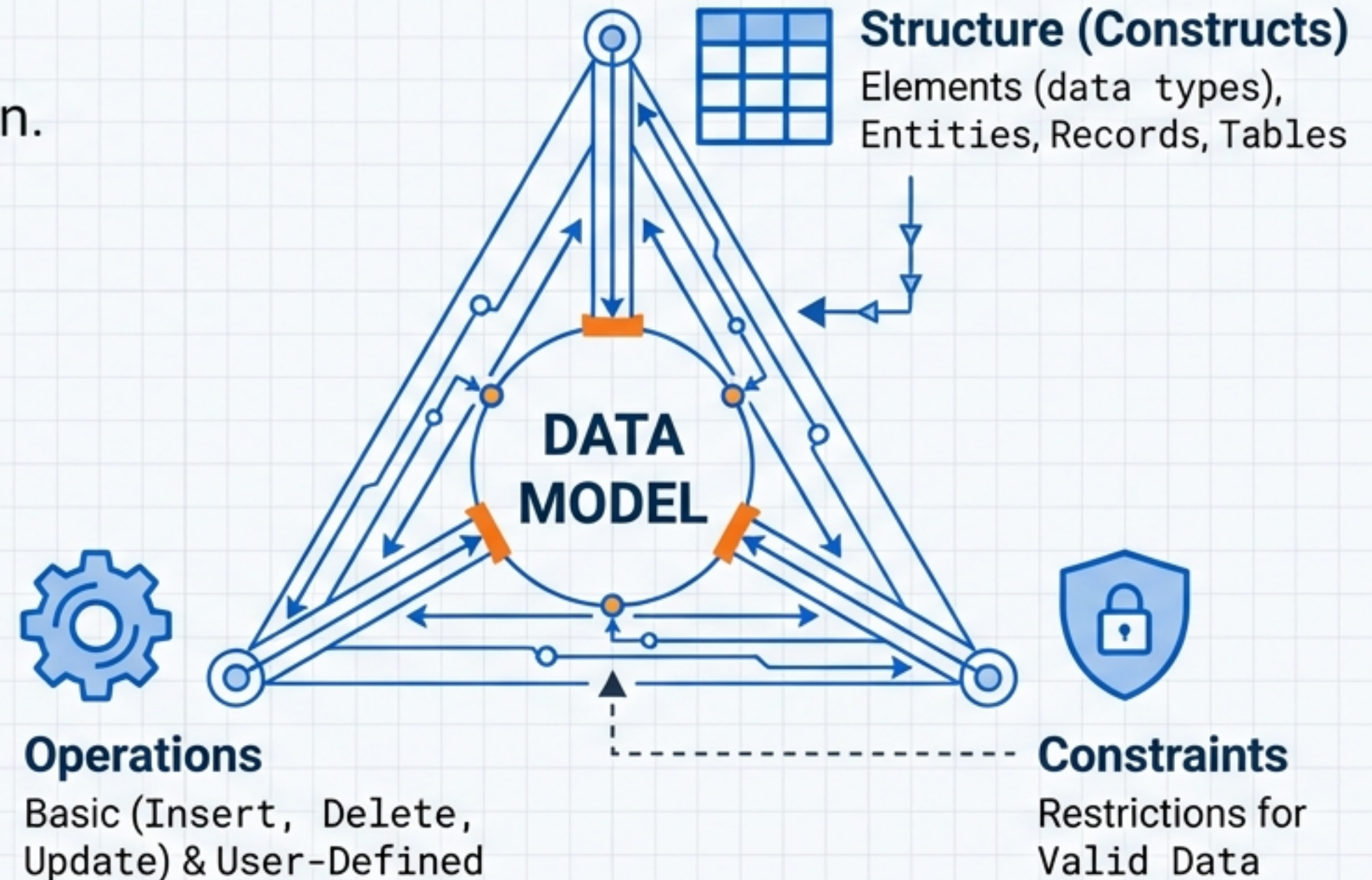


Source: Fundamentals of Database Systems, Elmasri & Navathe, Chapter 2

The Foundation: Data Models

A data model is a collection of concepts used to describe the structure of a database. It provides the necessary means to achieve data abstraction.

The Three Pillars of a Data Model



Categories of Data Models: A Spectrum of Abstraction

User View
(High Abstraction)

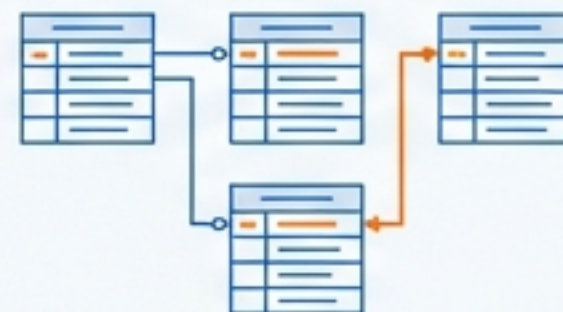
Conceptual Data Models

Close to user perception.
Uses Entities, Attributes, Relationships.
Example: ER Model.



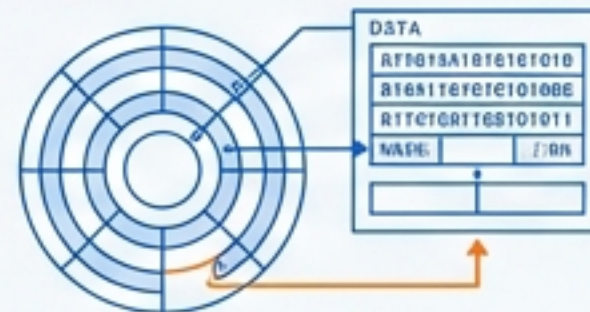
Representational (Implementation) Models

The bridge. Hides storage details but allows direct implementation.
Example: Relational Model (Tables).



Physical Data Models

Describes storage details.
Record formats, orderings, access paths.



Computer Storage (Low Abstraction)

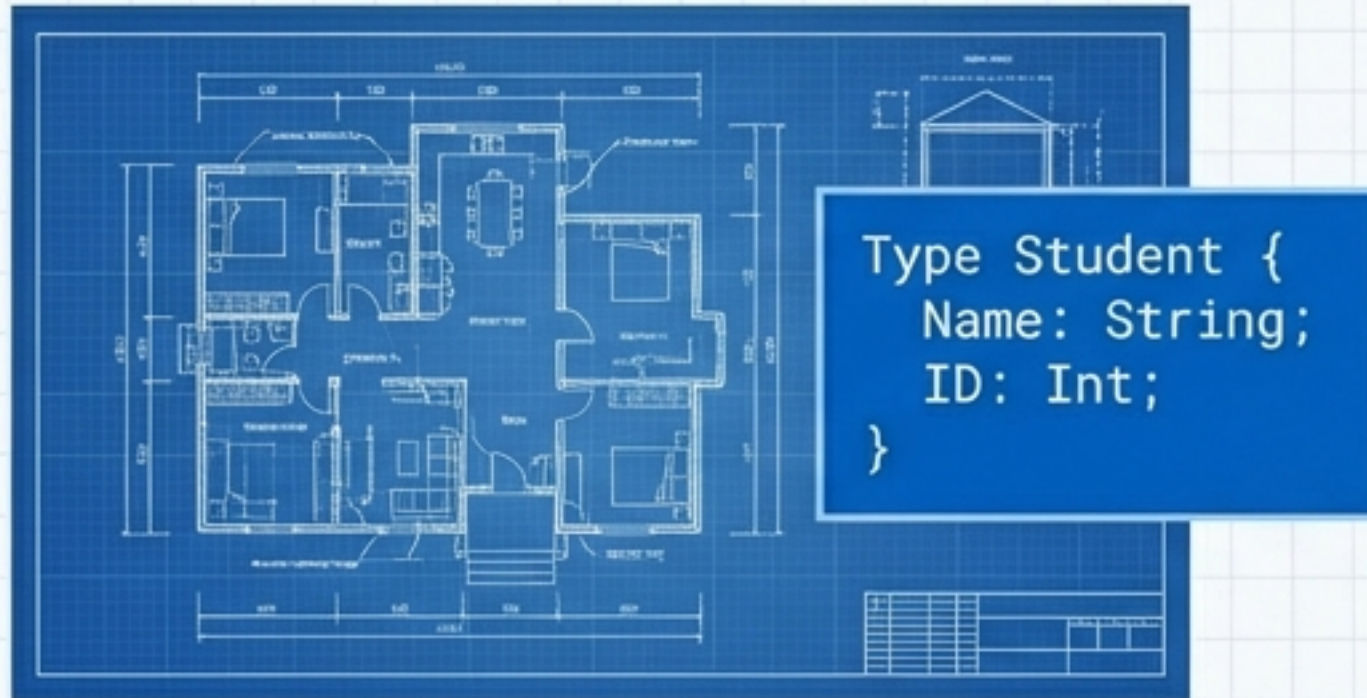
The Modern Outlier: Self-Describing Models

Combines description with data values (e.g., XML, NoSQL, Key-Value).

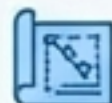


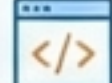
The Critical Distinction: Schema vs. State

Schema (Intension)



The description of the database structure, types, and constraints.

 **Characteristic:** Specified during design. Changes infrequently.

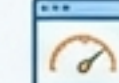
 **Analogy:** Variable definitions in code.

State (Extension)

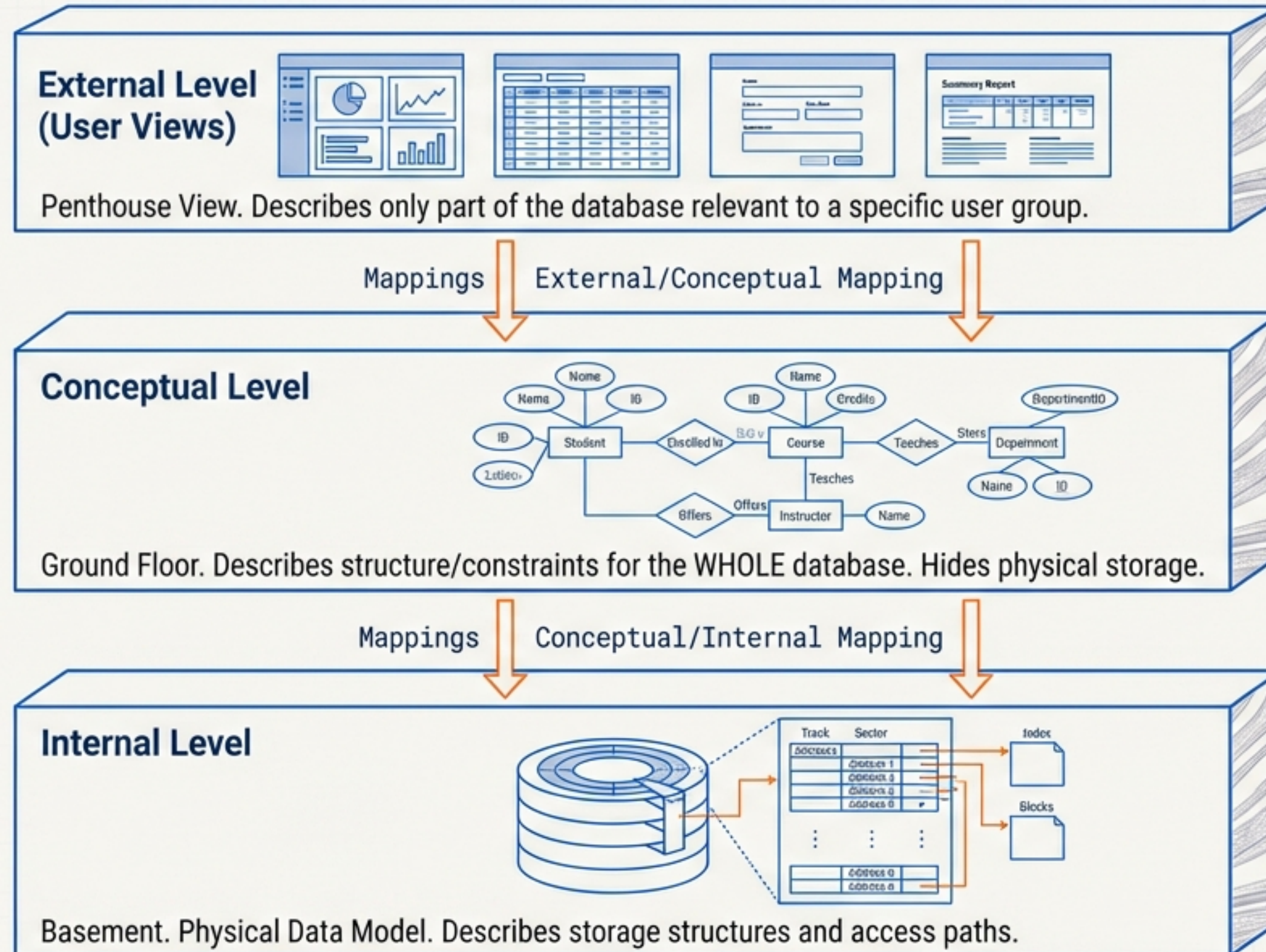


The actual data stored at a particular moment (Snapshot/Instance).

 **Characteristic:** Changes constantly with every update. Must remain a 'Valid State'.

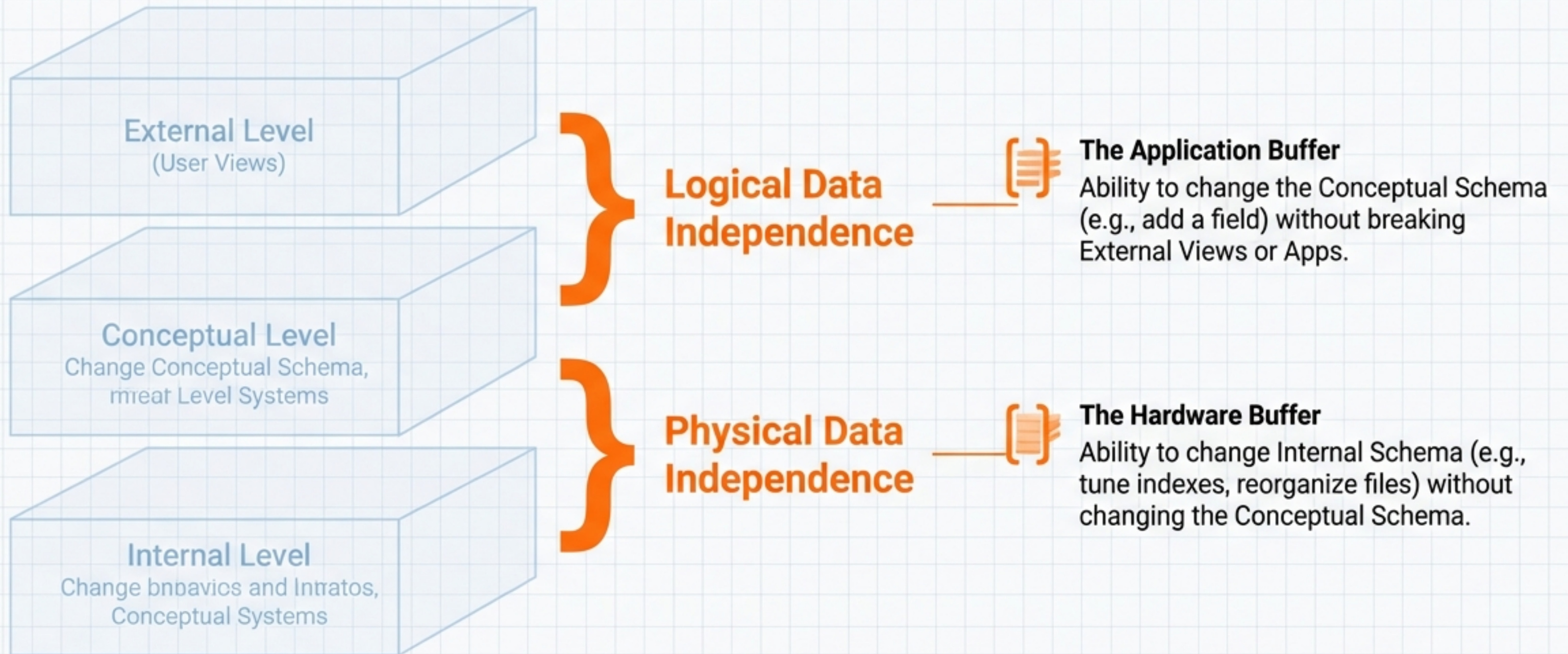
 **Analogy:** Values of variables at runtime.

The Three-Schema Architecture

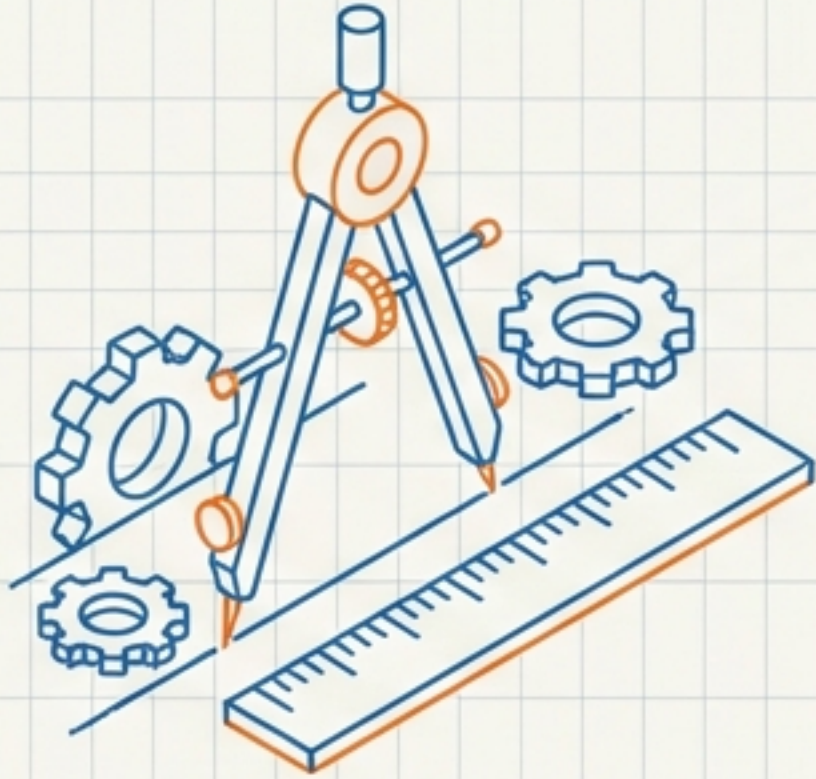


The Goal: Data Independence

Insulating the upper layers from changes in the lower layers.

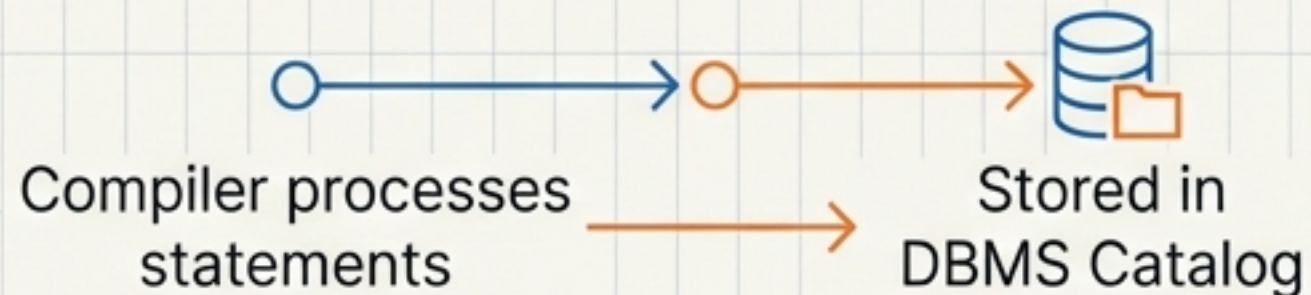


Speaking the Language: DBMS Languages



DDL: Data Definition Language

Used by DBAs/Designers to define Schemas.



DML: Data Manipulation Language

Used for CRUD operations (Retrieval, Insertion, Deletion, Modification).

- **High-Level (Declarative):** Set-oriented, 'What to get' (e.g., SQL).
- **Low-Level (Procedural):** Record-at-a-time, 'How to get it' (Looping).

Interfaces & Actors



Casual / Naive Users → Menu-Based / Mobile Apps (Browse & Touch)



Parametric Users → Forms-Based (Canned Transactions)



Sophisticated Users → Query Languages (SQL)

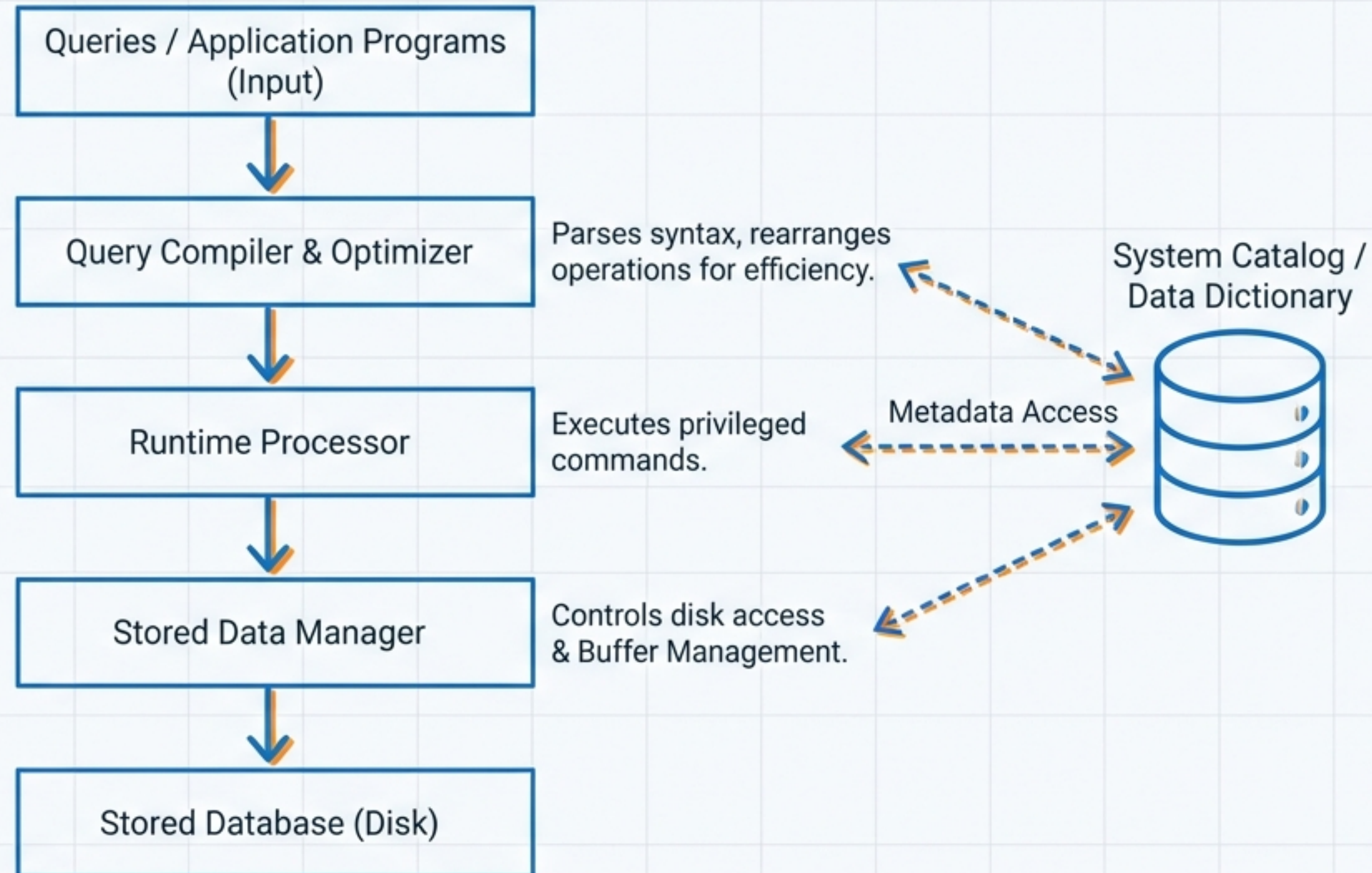


Database Administrators (DBAs) → Admin Tools (Privileged commands, Tuning)



Application Programmers → App Libraries (Embedded DML calls)

Under the Hood: DBMS Component Modules



The Support Ecosystem: Utilities & Tools

DBA Utilities



— Loading: Importing data files.



— Backup: Disaster recovery copies.

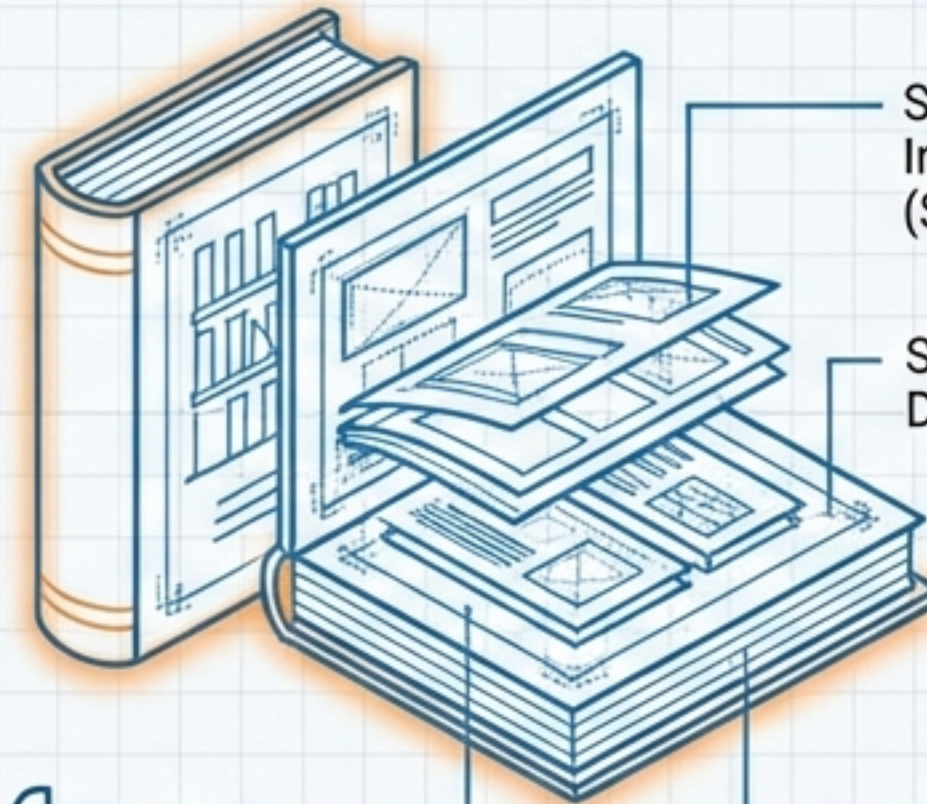


— File Reorganization:
Reindexing for performance.



— Performance Monitoring:
Usage statistics & tuning.

Data Dictionary (Repository)



Stores Catalog
Information
(Schemas, Constraints)

Stores Design
Decisions & Standards



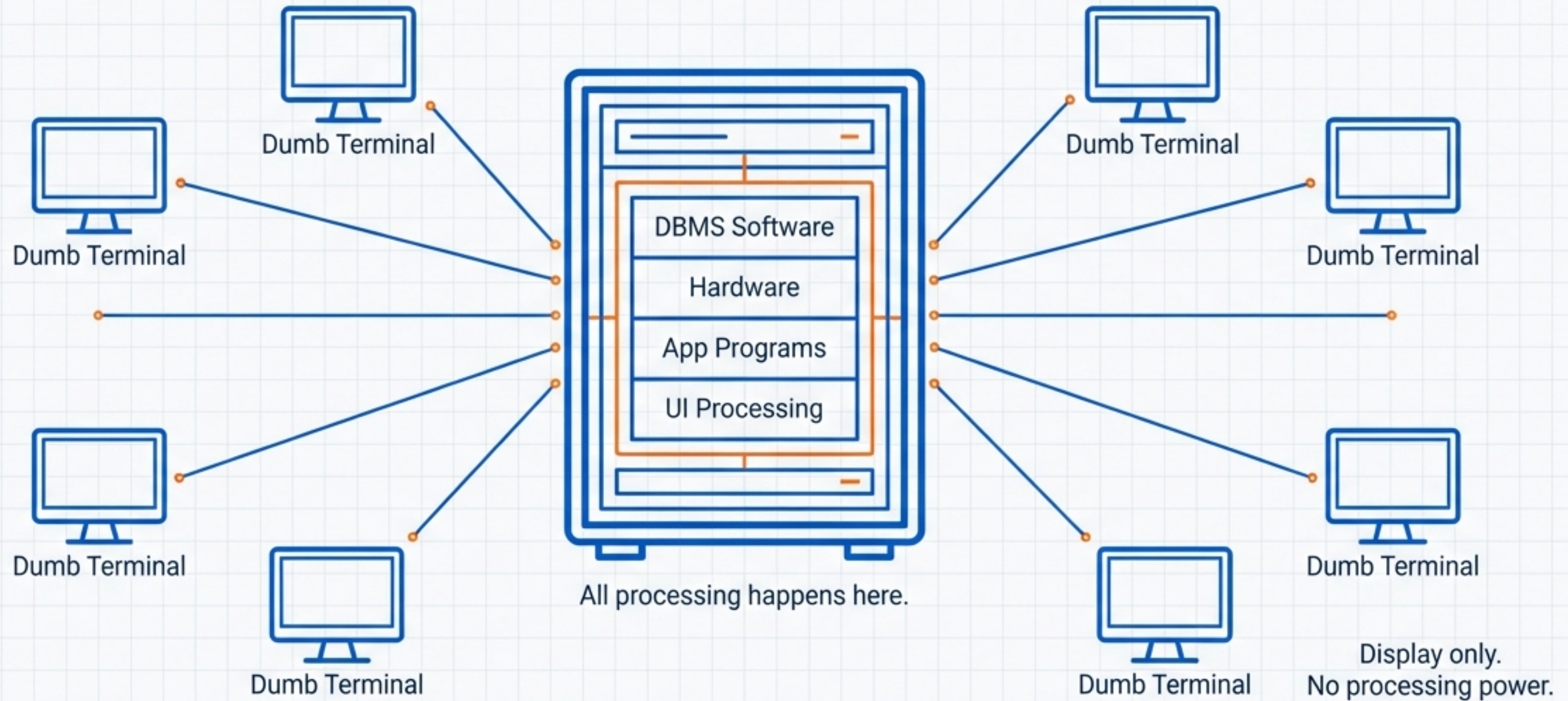
Active Repositories:
Accessed by DBMS software.



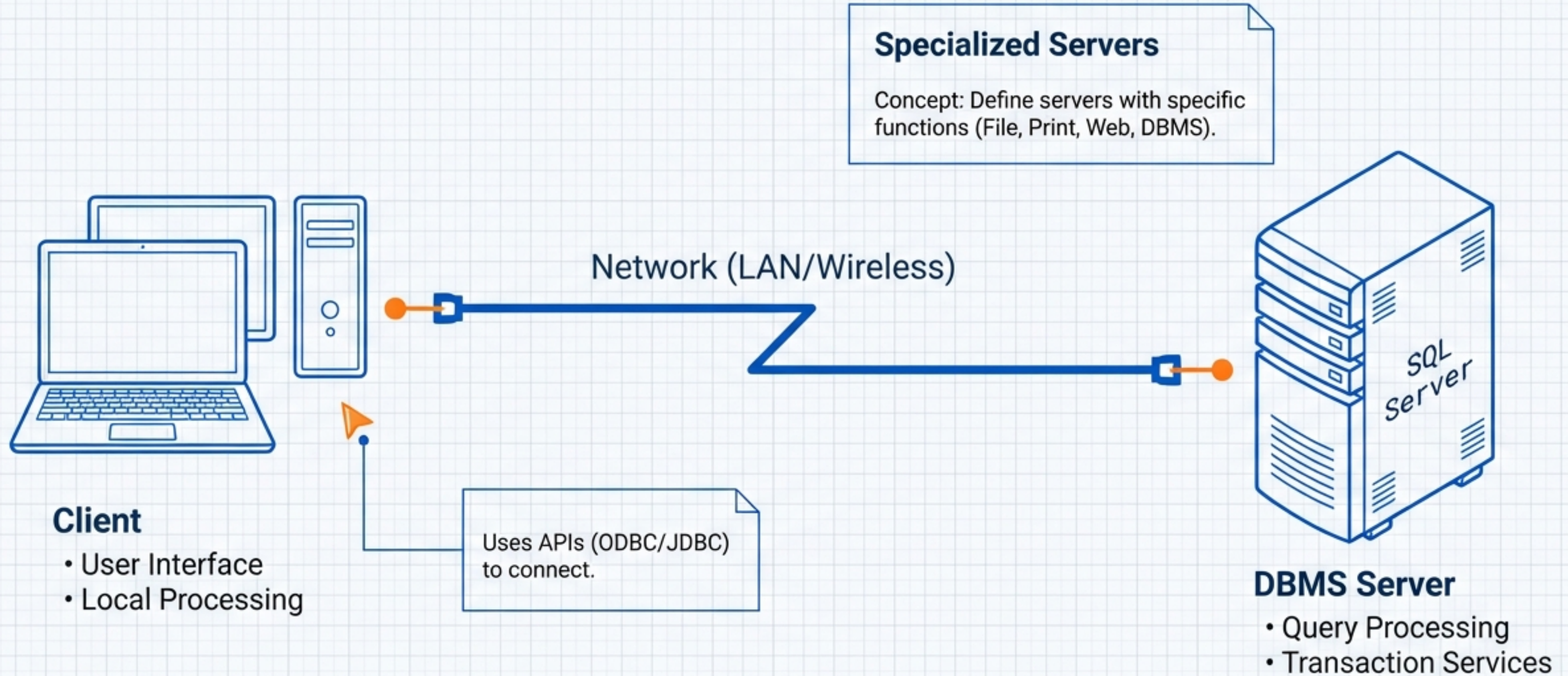
Passive Repositories:
For human documentation.

Physical Architecture: The Centralized Era

Helvetica Now Display, Deep Engineering Blue (#003366)



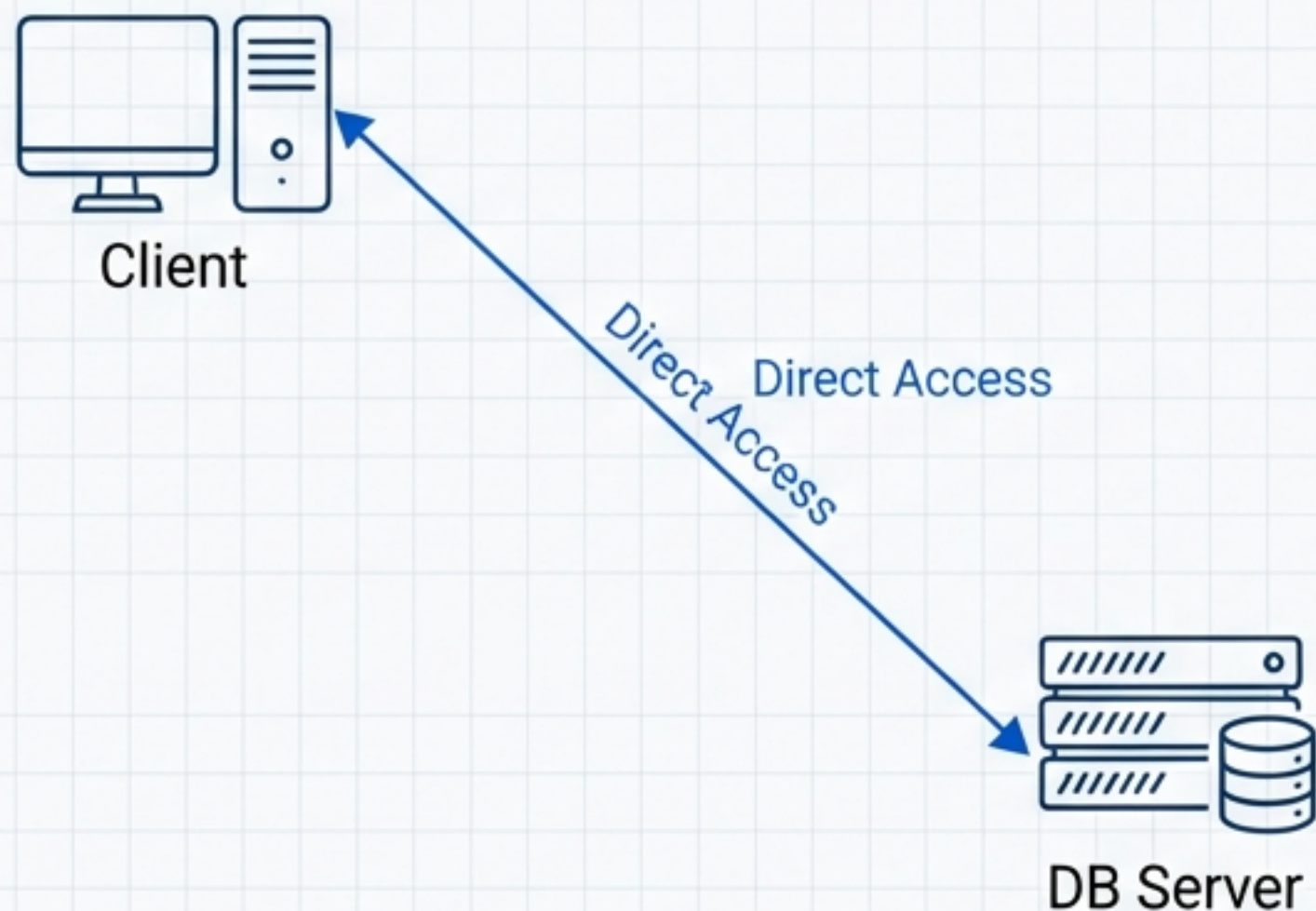
Evolution: Basic Client-Server Architecture



The Modern Standard: Two-Tier vs. Three-Tier

Two-Tier (Thick Client)

Technical Slate (#374151) Roboto



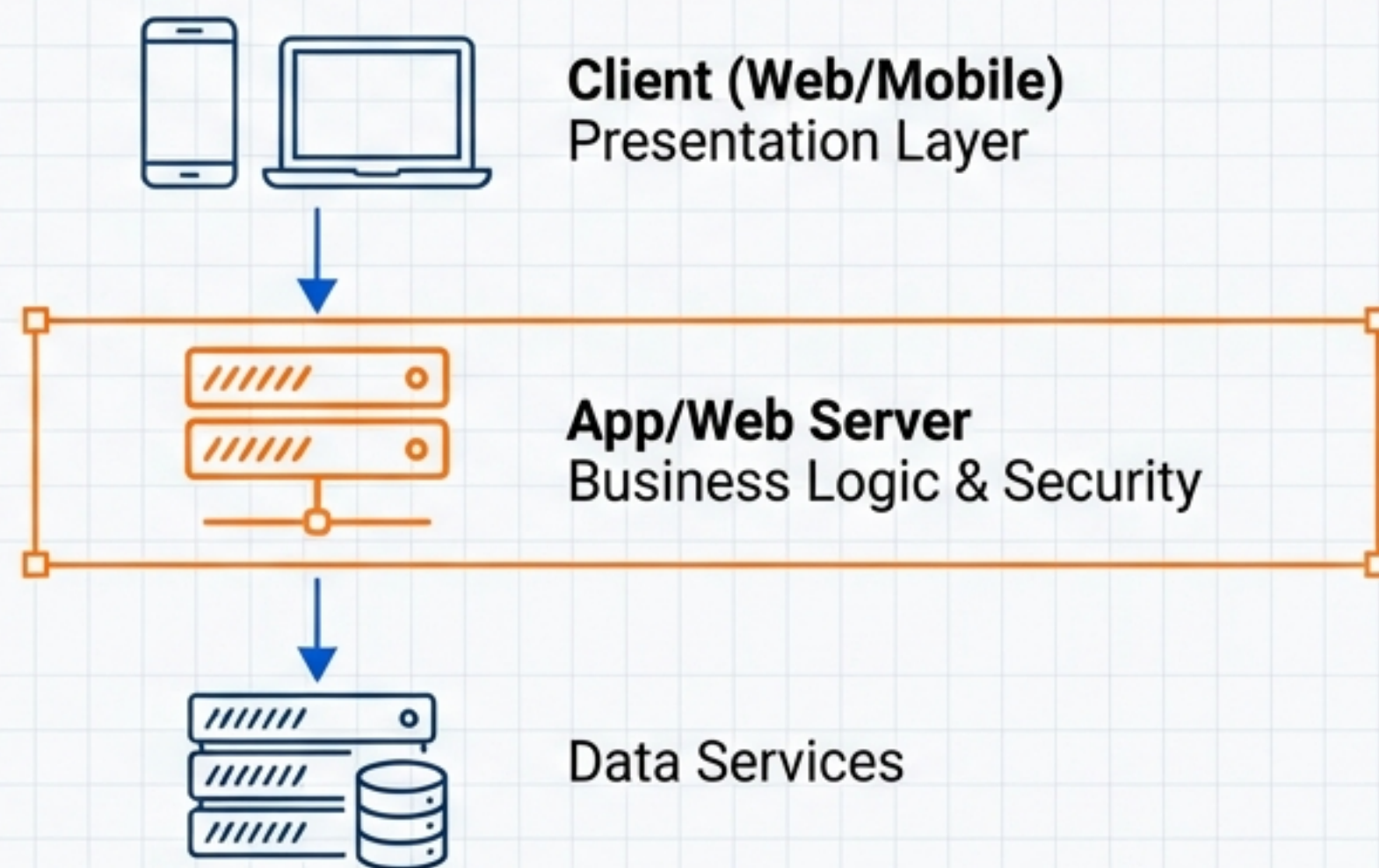
Notes in Roboto

Issues: Security risk, Scalability limits.

Roboto Mono: Off-White

Three-Tier (Web Architecture)

Technical Slate Roboto



Notes in Roboto

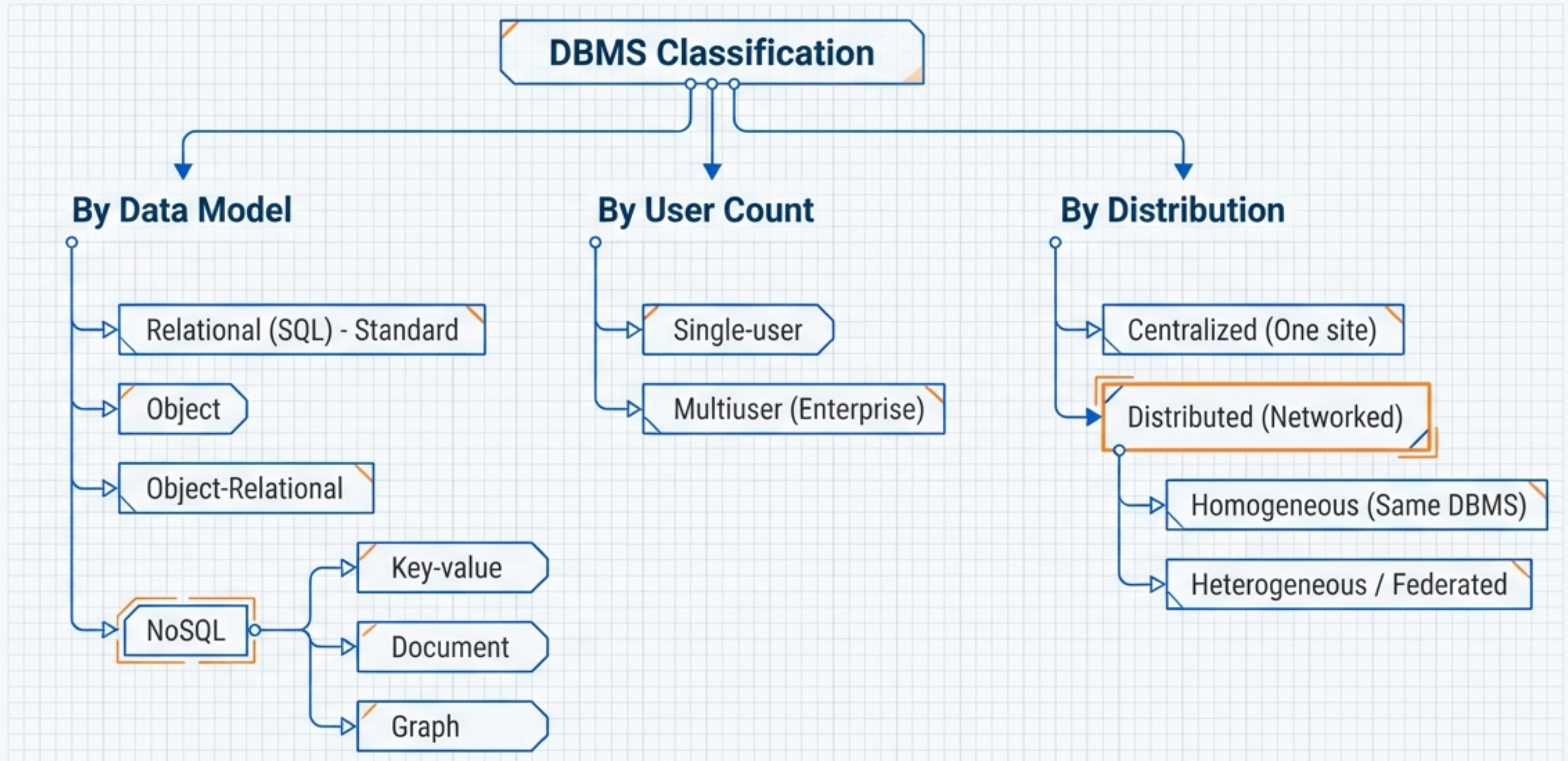
Benefits: Enhanced security, Decoupling.

Roboto Mono: Off-White

Classification of DBMSs

Concept

Diagram



Summary: The Architectural Blueprint

PROJECT CHECKLIST & SPECIFICATIONS

- ✓ **1. Abstraction**
Data Models define the grammar, separating concept from storage.
- ✓ **2. Insulation**
Three-Schema Architecture & Data Independence protect the system from change.
- ✓ **3. Translation**
DBMS modules convert high-level queries into low-level **disk ops**.
- ✓ **4. Deployment**
Evolution from **Mainframes** to **Three-Tier Web Standards** balances security and scale.

